**The Chinese University of Hong Kong, Shenzhen**
**SDS · School of Data Science**

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# MAT 3007 — Optimization

Exercise Sheet 2

**Problem 1 (Multiple Choice – Polyhedra and LPs):** (approx. *20* pts)
Decide whether the following statements are true or false. Please explain your answer and give suitable (counter)examples.

a) The intersection of two nonempty convex sets $S_1, S_2 \subseteq \mathbb{R}^n$ is again a convex set.

b) The union of two polyhedra $P_1, P_2 \subseteq \mathbb{R}^n$ is a polyhedron.

We now consider an LP in its standard form:

$$\min_{\boldsymbol{x}} \ \boldsymbol{c}^\top \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}, \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$ are given. Again decide whether the next statements are true or false and provide suitable explanations.

c) The set of all optimal solutions (assuming existence) of (1) must be bounded.

d) If $\boldsymbol{A}$ has full row rank, i.e., if the rows of $\boldsymbol{A}$ are linearly independent, then the LP (1) has a unique global solution.

e) If there is more than one optimal solution, then there are infinitely many optimal solutions.

**Problem 2 (LPs and Standard Form):** (approx. *20* pts)
Reformulate the problem

$$\begin{array}{lrl} \text{minimize} & |x_1 + x_2 + x_3| - x_2 & \\ \text{subject to} & (x_1 - 1)^2 & \leq 4 \\ & \max\{|x_2 + 1| - 2x_3, x_3\} & \leq 2 \\ & x_1 - x_2 - x_3 & \geq 0 \end{array}$$

as a linear optimization problem and write down its standard form.

**Problem 3 (Graphical Solution):** (approx. *15* pts)
Solve the following two-dimensional linear optimization problem using the graphical method as in the lecture:

$$\begin{array}{lrl} \text{maximize} & 2x_1 + 3x_2 & \\ \text{subject to} & 3x_1 - x_2 & \geq 0 \\ & x_1 - 4x_2 & \leq 0 \\ & 5x_1 - 2x_2 & \leq 18 \\ & x_1 + 2x_2 & \leq 18 \\ & 4x_1 - 3x_2 & \geq -5 \end{array}$$

Which constraints are active at optimal solution? Also list all the vertices of the feasible region.

**Problem 4 (Inscribing Squares):** (approx. *25* pts)

Let $P$ be a polyhedron defined by the following extreme points:

$$\begin{bmatrix}0\\8\end{bmatrix}, \quad \begin{bmatrix}2\\0\end{bmatrix}, \quad \begin{bmatrix}10\\3\end{bmatrix}, \quad \begin{bmatrix}12\\10\end{bmatrix}, \quad \begin{bmatrix}7\\12\end{bmatrix}, \quad \begin{bmatrix}3\\12\end{bmatrix}.$$

In this assignment, we want to find a square with the largest area that is fully contained in the polyhedron $P$.

a) Formulate a linear optimization program to solve this problem under the assumption that the square is aligned with the $x$- and $y$-axis (i.e., we do not consider rotations of the square).

   Explain the different components of your model and briefly discuss the correctness of your formulation.

b) Solve the described problem using `MATLAB` or `Python` and visualize your result.

**Problem 5 (Sparse Robust Regression):** (approx. *20* pts)

In this exercise, we consider the following optimization problem:

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \sum_{i=1}^{m} |\boldsymbol{a}_i^\top \boldsymbol{x} - b_i| \qquad \text{s.t.} \qquad \|\boldsymbol{x}\|_1 \leq \tau, \tag{2}$$

where $(\boldsymbol{a}_i, b_i)$, $i = 1, \ldots, m$ are given data points with $\boldsymbol{a}_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$ for all $i$, $\tau > 0$ is a parameter, and $\|\boldsymbol{y}\|_1 = \sum_{j=1}^{n} |y_j|$ denotes the $\ell_1$-norm of a vector $\boldsymbol{y} \in \mathbb{R}^n$.

This model is connected to data fitting and regression problems which are common tasks in statistics and machine learning. The underlying motivation is to model a linear relationship between $\boldsymbol{a}_i$ and $b_i$, i.e.,

$$b_i \approx \boldsymbol{a}_i^\top \boldsymbol{x}, \quad \forall\, i = 1, \ldots, m.$$

Here, $\boldsymbol{x} \in \mathbb{R}^n$ is the (unknown) parameter of the linear relationship. The optimization model (2) aims to determine $\boldsymbol{x} \in \mathbb{R}^n$ in this linear relationship that best fits the given data by minimizing the absolute residual errors $|\boldsymbol{a}_i^\top \boldsymbol{x} - b_i|$, $i = 1, \ldots, m$. In addition, we require $\boldsymbol{x} \in \mathbb{R}^n$ to have a certain "sparse" structure by imposing $\|\boldsymbol{x}\|_1 \leq \tau$ (the components of $\boldsymbol{x}$ should be small).

a) Rewrite the optimization problem (2) as a linear problem and explain your different reformulation steps in detail.

b) We now consider a special "linear fitting" instance of (2) with dimension $n = 2$. Specifically, given the data points $(u_i, v_i)^\top \in \mathbb{R}^2$, $i = 1, \ldots, m$, we want to find $\boldsymbol{x} = (x_1, x_2)^\top$ such that

$$v_i \approx x_1 u_i + x_2 \quad \forall\, i = 1, \ldots, m.$$

   Here, the data points $(u_i, v_i)^\top$ are selected from the data set

$$\mathcal{D} := \left\{ \begin{bmatrix}0\\8\end{bmatrix}, \begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}10\\3\end{bmatrix}, \begin{bmatrix}12\\10\end{bmatrix}, \begin{bmatrix}7\\12\end{bmatrix}, \begin{bmatrix}3\\12\end{bmatrix} \right\}, \quad \text{(i.e., we have } m = 6\text{)}.$$

   Write this task in the form (2), i.e., specify $\boldsymbol{a}_i$ and $b_i$. Solve the resulting regression problem using `MATLAB` or `Python` for $\tau \in \{0.1, 1, 10\}$ and visualize the obtained results.

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Sheet 2 is due on **Feb. 24th**. Submit your solutions before **Feb. 24th**, **11:00 pm**.